The box

Above: ARCHITECTURE.md §3 (the box around every program) and §8 (running it). **Talks to:** every program on it. The front door is its only open port; the backups carry its files off it.

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1. What it is

The one server everything runs on, and how it is run. It is a small cloud server with 1 GB of memory. One supervisor keeps every program running inside its own memory budget, one command deploys new code, and the keys live in protected files that never leave it.

The machine	
Machine	AWS EC2 **t3.micro**: 2 vCPU, **912 MiB** of memory
Swap	a **2 GB swap file**, made once by `deploy/add-swap.sh`; the deploy requires it
System	Amazon Linux 2023. Its `python3` is 3.9: the Python programs are tested on 3.9
User and folder	`ec2-user`, `~/Tade-desk-server` (one checkout of the code repository)

What it does:

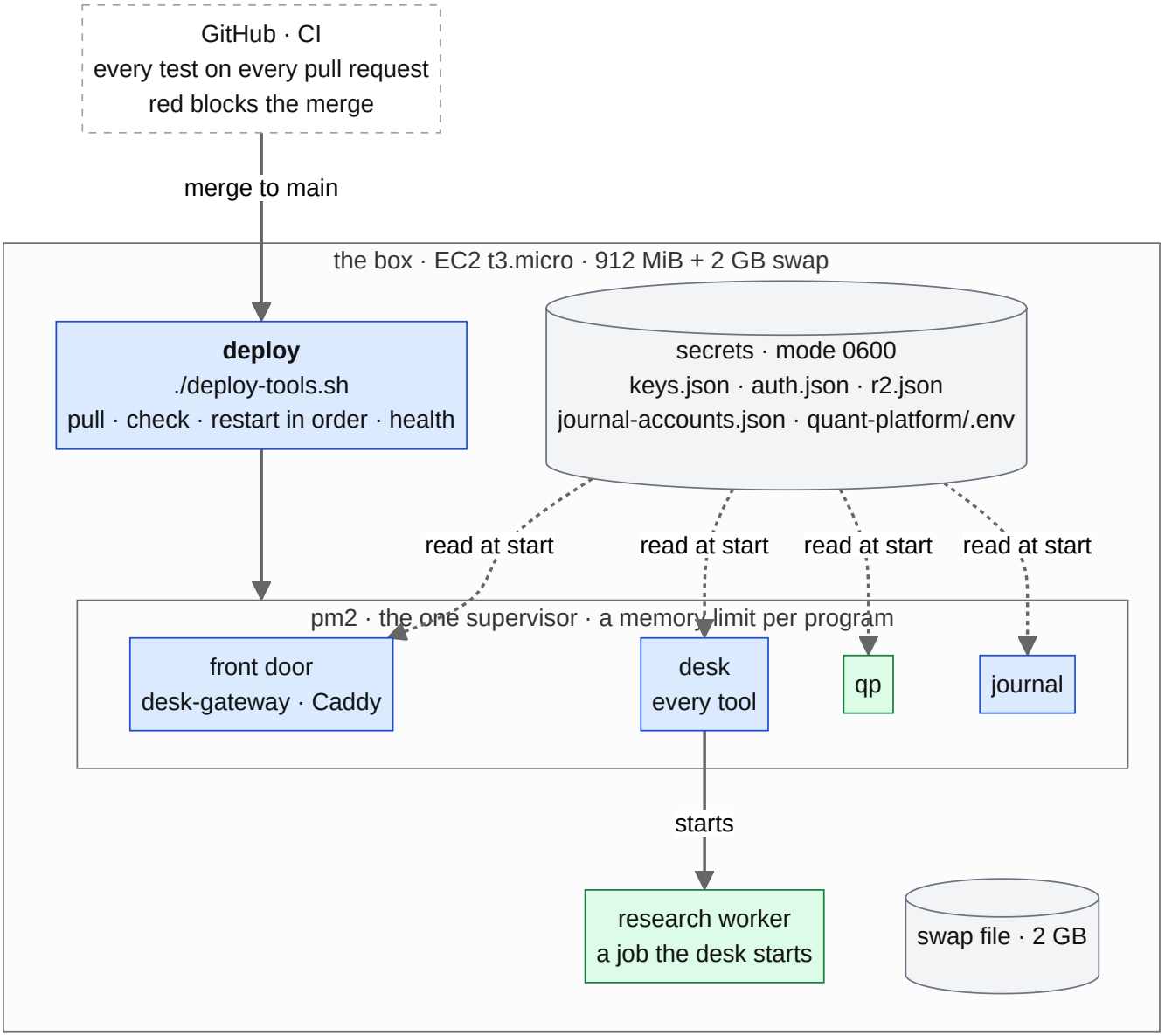
- runs every program of the desk under **one supervisor** (pm2), each with a memory limit;
- deploys the code with **one command**;
- keeps the secrets in files only its owner can read, which never leave the box.

What it never does:

- run a program under two supervisors;
- keep a secret in git, in a log or in a backup;
- accept a deploy in the minutes around a photo (once built, §4.4).

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2. The diagram



3. The programs and their memory

Measured on 2026-10-06, today's arrangement: about 724 MB of programs, beside the system.

Program	Memory
five tool programs	89–91 MB each
qp	129 MB
the archive	78 MB
the journal	66 MB

Program	pm2 name	Port (local)	Memory budget	pm2 limit
desk (every tool)	`desk`	3000 (home), and one per tool: 3010 … 3120; T1 on 3090 once it moves (WEB.md §3.1)	about 150–250 MB (estimate; measured as tools move in)	to set from the measurement

Program	pm2 name	Port (local)	Memory budget	pm2 limit
qp	`qp`	8765	70–150 MB (129 measured)	`deploy/qp.config.js`
journal	`journal`	3100	about 66 MB	160 MB
front door	`desk-gateway`, Caddy	8080; 443 outside	40–80 MB	80 MB (gateway)
research worker	— (a job)	—	100–150 MB for new tickers; 200–400 MB at night	—
system	—	—	about 150 MB	—

About 500–650 MB in all. The night's research run happens at 18:30, when no scan is due, and may use swap. A 2 GB instance removes the squeeze; that is the trader's decision (ARCHITECTURE §10).

Today's limits, until the tools move into the desk:

Program	pm2 limit
T1	280 MB
T2, T6	240 MB
T11	200 MB
T7	180 MB
any other tool	140 MB
the archive	120 MB
the desk's comparison copy	300 MB

pm2 restarts a program that passes its limit. Nothing is started by Manage tools with less than 180 MB available, because on a 912 MiB box the alternative is the kernel choosing what to kill, and it may choose `sshd`.

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4. Its parts

4.1 The supervisor

pm2 for every program of the desk.

- Every program of the desk is a pm2 process with a memory limit and timestamps in its log.
- `pm2 save` and `pm2 startup` bring them back after a reboot.

Each program has exactly one supervisor, never two. Two supervisors once restarted one program 113,854 times (2026-09-23). The box's own services (sshd; Caddy and the DuckDNS updater, FRONT-DOOR.md) stay under systemd and cron as system services, never also under pm2.

The one exception, to be retired: a systemd timer, `qp-daily.timer`, still runs qp's nightly O'Neil chain. `scripts/retire-old-schedulers.sh` removes it and the pm2 backup cron. The desk's night job then runs the chain and the backup (JOBS.md).

4.2 The deploy

One command on the box: `./deploy-tools.sh`.

It	Why
pulls itself first, and runs itself again if the script changed (guarded against looping)	bash keeps reading the old file, so a change used to take effect one deploy late
reads the tool registry **after** the pull	or it starts the previous deploy's tools
refuses to reset when the branch would take `quant-platform/` back to an older state	a deploy from an older branch once silently reverted qp
installs what each program needs (`npm ci`; `psycopg2-binary` for the research worker when missing)	
backs up every file and the shared record before it restarts anything (to build, BACKUPS.md)	a restart is when migrations run, so there is always a copy from just before a schema changed
restarts in order: the desk (today: each tool through `deploy/tool-ctl.sh`), the comparison copy when one is listed, the archive (until it retires), the journal, the front door, qp (through `deploy/qp-restart.sh`, only if qp is running older code)	
copies the Alpaca pair into `quant-platform/.env` (`scripts/sync-qp-env.js`)	so qp and the desk never hold different keys
checks every program's `/health`, and what each data feed actually returned (`/api/feedcheck`, fetched, not inventoried)	a key that is present is not proof that it works

To build. The deploy refuses to run within 15 minutes of any tool's capture minute on a trading day, unless forced. A deploy at a photo minute costs that tool a day of research.

CI, before the deploy. Every pull request runs every test: Node (Jest) and Python (qp's `run_all`, the research worker), against a real Postgres. Red blocks the merge.

4.3 The secrets

File	Holds	Set with
`data/keys.json`	the shared keys and settings of the desk: `databaseUrl`, `alertNtfyTopic`, the area switches (`factsMode`, `marketMode`, ...), `workerShadow`	`node scripts/set-setting.js <name>`: asks with echo off, checks a database address before saving
`data/auth.json`	the password hash and the cookie's signing secret	`scripts/set-password.sh`
`data/r2.json`	the backup bucket's keys	`scripts/set-r2-keys.sh`
`data/journal/journal-accounts.json`	the journal's Alpaca accounts	the journal's Accounts page
`quant-platform/.env`	qp's feed keys	copied by the deploy from the desk's keys

The rules:

- **Mode 0600**, gitignored, never printed, never logged, never backed up.
- The trader keeps a copy off the box; a restore puts the secrets back from that copy (BACKUPS.md).

- Secrets are entered on the box, never pasted into a chat or a log.

To build. One secret store, written by one command (`desk secret set <name>`) and read by every program, so that no key is copied between files.

4.4 Ports

Inside the box	Outside
each program keeps its own local port; programs call each other on `127.0.0.1`	**today:** the programs' ports are still open to the internet. **The design:** only 443 (the front door) is open

The login treats a call from the box itself as local only when it did not pass through the front door (FRONT-DOOR.md).

4.5 Logs

pm2 keeps each program's output, with timestamps. A pm2 restart count is the only trace a memory-limit kill leaves, so the deploy prints each program's restart count.

To build:

- rotated logs;
- the last errors of every program, shown on the health page (ALERTS.md).

4.6 The old helper scripts, and where each goes

Script today	Where it goes
`scripts/set-password.sh`, `set-r2-keys.sh`, `set-alpaca-keys.js`, `set-setting.js`, `sync-qp-env.js`	one secret store and `desk secret set` (§4.3)
`deploy-tools.sh`, `deploy/qp-restart.sh`, `deploy/add-swap.sh`	kept: the one deploy; swap is made once
`deploy/tool-ctl.sh`	retired as the tools move into the desk: a tool is a context, not a process. Manage tools adds it to the desk instead (HOME.md)
`show-screeners`, `check-screeners`, `why-empty`, `try-filter`, `field-values`, `verify-tv-fields`, `probe-*`	the screener builder's Test, the adapters' live column list, and the scheduled health checks (SCREENERS.md, DATA-SOURCES.md)
`audit-floor`, `purge-sub-floor`, `backfill_signals`, `rebuild_training`, `convert_legacy_to_r4.py`, `split-tool-history`, `rebuild-tool-db`	one-time repairs, kept as scripts until the import of the old data is complete (MIGRATION.md)
`analyse-bias`, `analyse-news-chain`, `card_lookup.py`, `verify_irdm_score.py`, `export-training.js`, `archive-values.js`	the register exports and the API: a card as of a day, the R1 register, the score's breakdown
`scripts/desk.js` (restore), `backup-r2.js`, `restore-r2.js`, `import-registers.js`, `facts-report.js`, `worker-report.js`	kept (BACKUPS.md, SHARED-RECORD.md, MIGRATION.md)
qp's `run_daily.py`, `run_backfill.py`, `run_edgar.py`, `run_13f.py`, `run_sic.py`, `run_status.py`, `run_fields.py`	the research worker's steps, started by the desk's scheduler; their checks on the health page (RESEARCH-WORKER.md)

Script today	Where it goes
qp's `tools/` (compare server, catalogue, strategy checks, recovery, rank sweep), `exec-compare.js`, `fill-compare.js`, `check-feeds.js`	kept with qp as they are (QP.md)

5. Contracts

It provides	To
a supervised process with a memory limit, restarted after a crash or a reboot	every program
one deploy command with health checks	the trader
the secret files, mode 0600	every program (read at start)
`127.0.0.1` between programs	every program
the 2 GB swap file	the night's research run

6. When it fails

Event	What happens
a program passes its memory limit	pm2 restarts it. The desk brings every tool's list back from its file; the restart count shows it
memory runs short	swap absorbs a peak. Manage tools refuses to add a tool with less than 180 MB available
a deploy fails midway	it stops at the failing step and says which. The programs already restarted keep running; the rest keep their previous code
the box reboots	pm2 brings every program back; jobs whose slot passed are recorded as missed and alert (JOBS.md)
the box is lost	a new box: install, put back the secrets from the trader's copy, then `node scripts/desk.js restore <date>` (BACKUPS.md). The shared record still holds the copy of every photo taken since the last backup

7. Built on

The old desk:

- today's architecture;
- the deploy script's own recorded lessons (§4.2);
- the code repository's `deploy/README.md`;
- the qp operations;
- the journal operations.

What failed there:

- **Two supervisors:** 113,854 restarts on 2026-09-23.
- **A qp restart took 3 min 20 s:** the port check refused the port while old connections lingered. Fixed on 2026-10-06.
- **Keys pushed to GitHub** by an old backup (S-2).
- **Tools stopped for lack of memory:** five tool programs at about 90 MB each.

- **No CI** until step 1.

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8. Status

Part	Status
the machine, swap, pm2, the deploy	**built**
CI	**built** (step 1)
the night handed to the desk's scheduler	**built, not switched**: `scripts/retire-old-schedulers.sh` is ready; the old timer still runs
the photo-window guard on the deploy	**to build** (migration step A)
one secret store	**to build**
only 443 open	**to do by the trader** once the domain is set (FRONT-DOOR.md)
the desk as one pm2 process for every tool	**being built** (DESK.md, MIGRATION.md steps A and B)